

## Coursework

It has 3 parts

- Practice SQL. (10 marks)
- Group project (group of 2 students) with discussions. (10 marks)
- Test. (10 marks)

## Part 1

Q1.

(a)

| EMPLOYEE |       |         |           |            |                          |     |        |           |     |  |
|----------|-------|---------|-----------|------------|--------------------------|-----|--------|-----------|-----|--|
| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                  | Sex | Salary | Super_ssn | Dno |  |
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M   | 30000  | 333445555 | 5   |  |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX    | M   | 40000  | 888665555 | 5   |  |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX  | F   | 25000  | 987654321 | 4   |  |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX  | F   | 43000  | 888665555 | 4   |  |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M   | 38000  | 333445555 | 5   |  |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX   | F   | 25000  | 333445555 | 5   |  |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX  | M   | 25000  | 987654321 | 4   |  |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX   | M   | 55000  | NULL      | 1   |  |

| DEPARTMENT     |         |           |                |
|----------------|---------|-----------|----------------|
| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
| Research       | 5       | 333445555 | 1988-05-22     |
| Administration | 4       | 987654321 | 1995-01-01     |
| Headquarters   | 1       | 888665555 | 1981-06-19     |

| DEPT_LOCATIONS |           |
|----------------|-----------|
| Dnumber        | Dlocation |
| 1              | Houston   |
| 4              | Stafford  |
| 5              | Bellaire  |
| 5              | Sugarland |
| 5              | Houston   |

| PROJECT         |         |           |      |
|-----------------|---------|-----------|------|
| Pname           | Pnumber | Plocation | Dnum |
| ProductX        | 1       | Bellaire  | 5    |
| ProductY        | 2       | Sugarland | 5    |
| ProductZ        | 3       | Houston   | 5    |
| Computerization | 10      | Stafford  | 4    |
| Reorganization  | 20      | Houston   | 1    |
| Newbenefits     | 30      | Stafford  | 4    |

| WORKS_ON  |     |       |
|-----------|-----|-------|
| Essn      | Pno | Hours |
| 123456789 | 1   | 32.5  |
| 123456789 | 2   | 7.5   |
| 666884444 | 3   | 40.0  |
| 453453453 | 1   | 20.0  |
| 453453453 | 2   | 20.0  |
| 333445555 | 2   | 10.0  |
| 333445555 | 3   | 10.0  |
| 333445555 | 10  | 10.0  |
| 333445555 | 20  | 10.0  |
| 999887777 | 30  | 30.0  |
| 999887777 | 10  | 10.0  |
| 987987987 | 10  | 35.0  |
| 987987987 | 30  | 5.0   |
| 987654321 | 30  | 20.0  |
| 987654321 | 20  | 15.0  |
| 888665555 | 20  | NULL  |

| DEPENDENT |                |     |            |              |
|-----------|----------------|-----|------------|--------------|
| Essn      | Dependent_name | Sex | Bdate      | Relationship |
| 333445555 | Alice          | F   | 1986-04-05 | Daughter     |
| 333445555 | Theodore       | M   | 1983-10-25 | Son          |
| 333445555 | Joy            | F   | 1958-05-03 | Spouse       |
| 987654321 | Abner          | M   | 1942-02-28 | Spouse       |
| 123456789 | Michael        | M   | 1988-01-04 | Son          |
| 123456789 | Alice          | F   | 1988-12-30 | Daughter     |
| 123456789 | Elizabeth      | F   | 1967-05-05 | Spouse       |

(b)

Figure 1.

1. Specify the following queries on the database shown in the Figure 1 (a) in SQL. Show the query results if each query is applied to the database state in figure (b).
  - a. For each department whose average employee salary is more than \$30,000, retrieve the department name and the number of employees working for that department.
  - b. Suppose that we want the number of male employees in each department making more than \$30,000, rather than all employees. Can we specify this query in SQL? Why or why not?

2. Consider the following view, DEPT\_SUMMARY, defined on the COMPANY database in Figure 1:

```
CREATE VIEW DEPT_SUMMARY (D, C, Total_s, Average_s)
AS SELECT Dno, COUNT(*), SUM (Salary), AVG (Salary)
FROM EMPLOYEE
GROUP BY Dno;
```

Note: MySQL has a configuration setting called IGNORE\_SPACE. If it's disabled (the default), COUNT (\*) is invalid, while COUNT(\*) is valid.

State which of the following queries and updates would be allowed on the view. If a query or update would be allowed, show what the corresponding query or update on the base relations would look like, and give its result when applied to the database in Figure 1.

- a. **SELECT \***  
**FROM DEPT\_SUMMARY;**
- b. **SELECT D, C**  
**FROM DEPT\_SUMMARY**  
**WHERE TOTAL\_S > 100000;**
- c. **SELECT D, AVERAGE\_S**  
**FROM DEPT\_SUMMARY**  
**WHERE C > ( SELECT C FROM DEPT\_SUMMARY WHERE D = 4);**
- d. **UPDATE DEPT\_SUMMARY**  
**SET D = 3**  
**WHERE D = 4;**
- e. **DELETE FROM DEPT\_SUMMARY**  
**WHERE C > 4;**

Q2.

**STUDENT**

| Name  | Student_number | Class | Major |
|-------|----------------|-------|-------|
| Smith | 17             | 1     | CS    |
| Brown | 8              | 2     | CS    |

**COURSE**

| Course_name               | Course_number | Credit_hours | Department |
|---------------------------|---------------|--------------|------------|
| Intro to Computer Science | CS1310        | 4            | CS         |
| Data Structures           | CS3320        | 4            | CS         |
| Discrete Mathematics      | MATH2410      | 3            | MATH       |
| Database                  | CS3380        | 3            | CS         |

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|--------------------|---------------|----------|------|------------|
| 85                 | MATH2410      | Fall     | 07   | King       |
| 92                 | CS1310        | Fall     | 07   | Anderson   |
| 102                | CS3320        | Spring   | 08   | Knuth      |
| 112                | MATH2410      | Fall     | 08   | Chang      |
| 119                | CS1310        | Fall     | 08   | Anderson   |
| 135                | CS3380        | Fall     | 08   | Stone      |

**GRADE REPORT**

| Student_number | Section_identifier | Grade |
|----------------|--------------------|-------|
| 17             | 112                | B     |
| 17             | 119                | C     |
| 8              | 85                 | A     |
| 8              | 92                 | A     |
| 8              | 102                | B     |
| 8              | 135                | A     |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---------------|---------------------|
| CS3380        | CS3320              |
| CS3380        | MATH2410            |
| CS3320        | CS1310              |

Figure 2.

1. Specify the following queries in SQL on the database schema in the Figure 2.
  - a. Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).
  - b. Retrieve the names and major departments of all students who do not have a grade of A in any of their courses.

## **Part 2**

### **Group project (two students)**

Building the back-end of a simple e-commerce system. Choose **either** of the following:

- Write the stored procedure in MySQL.
- Python programming to complete the transaction.py file.

The folders containing the files and what to do are provided.

Download the tools required depending on which one you need as explained next.

Create a database for an e-Commerce website. Assume a simple website where users can order items and pay from a “gift” card whose balance is managed by the website. Consider the following when designing your database

1. What entities do you need?
  2. What attributes do you need for these entities?
  3. What are the relations and their type between the entities.
  4. What are the primary keys?
  5. What are the foreign keys?
- 
1. Study the provided files and complete the TODO parts in SQL or python where required until you have a working application. Submit the files by email to [sarah.saeed3141@gmail.com](mailto:sarah.saeed3141@gmail.com).
  2. Draw the ER diagram for this database system.

### **Tools**

- SQLite
- MySQL workbench/PostgreSQL
- VS Code

## Downloads

### SQLite

SQLite is often faster for simple SELECT queries and for read-heavy workloads because it does not have the network overhead of MySQL. MySQL is much faster for high-concurrency writes and complex, optimized queries.

Note: SQLite does not support stored procedures in the traditional sense. So will practice stored procedures with MySQL or PostgreSQL.

### Download SQLite3

Go to <https://www.sqlite.org/download.html> and download the suitable package, for example **sqlite-tools-win-x64-3530000.zip**

Note: ensure to add the path to the windows environment variable so that you can use it from the windows command line.

```
cmd> where sqlite3
```

```
C:\Users\sarah\Downloads\sqlite-tools-win-x64-3530000\sqlite3.exe
```

### Documentation for SQLite3 for Python

<https://docs.python.org/3/library/sqlite3.html>

The image is a screenshot of a web browser displaying the Python documentation for the sqlite3 module. The browser's address bar shows the URL 'docs.python.org/3/library/sqlite3.html'. The page has a light blue header with the title 'sqlite3 — DB-API 2.0 interface for SQLite databases'. On the left side, there is a 'Table of Contents' sidebar with a tree view showing the module's structure, including sections like 'Tutorial', 'Reference', and 'How-to guides'. The main content area on the right contains the title 'sqlite3 — DB-API 2.0 interface for SQLite databases', the 'Source code' link, and a detailed description of the module. It explains that SQLite is a C library providing a lightweight disk-based database. It also mentions that the module was written by Gerhard Häring and is compliant with the DB-API 2.0 specification. A 'See also' section at the bottom points to the SQLite website. The page is styled with a clean, professional layout typical of official documentation.

**Ensure** to run the command line below on Windows command prompt to create the database

```
sqlite3 shop.db < schema.sql
```

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sarah>cd C:\Users\sarah\OneDrive\Desktop\DBCOURSEWORK\ecommerce_python_and_sql

C:\Users\sarah\OneDrive\Desktop\DBCOURSEWORK\ecommerce_python_and_sql>sqlite3 shop.db < schema.sql

C:\Users\sarah\OneDrive\Desktop\DBCOURSEWORK\ecommerce_python_and_sql>sqlite3 shop.db "SELECT * FROM products;"



| product_id | name                | description              | price  | stock |
|------------|---------------------|--------------------------|--------|-------|
| 1          | Laptop Pro 15       | High-performance laptop  | 999.99 | 10    |
| 2          | Wireless Mouse      | Ergonomic wireless mouse | 29.99  | 50    |
| 3          | USB-C Hub           | 7-in-1 USB-C hub         | 49.99  | 30    |
| 4          | Mechanical Keyboard | RGB mechanical keyboard  | 89.99  | 20    |
| 5          | Monitor 27"         | 4K IPS display           | 399.99 | 8     |
| 6          | Webcam HD           | 1080p webcam with mic    | 59.99  | 25    |



C:\Users\sarah\OneDrive\Desktop\DBCOURSEWORK\ecommerce_python_and_sql>
```

## Python with SQLite

...

Hints:

use `conn.execute()`, `conn.rollback()` with the appropriate SQLite syntax where the variables are passed to the function. For example:

(1) execute a query

retrieve a tuple

```
conn.execute(
    "SELECT * FROM tableX WHERE attributeY = ?", (attributeYvalue,)
).fetchone()
```

(2)

update a tuple

```
conn.execute(
    "UPDATE tableX SET attributeY = attributeY + ? WHERE attributeZ = ?",
    (attributeYvalue, attributeZvalue)
)
```

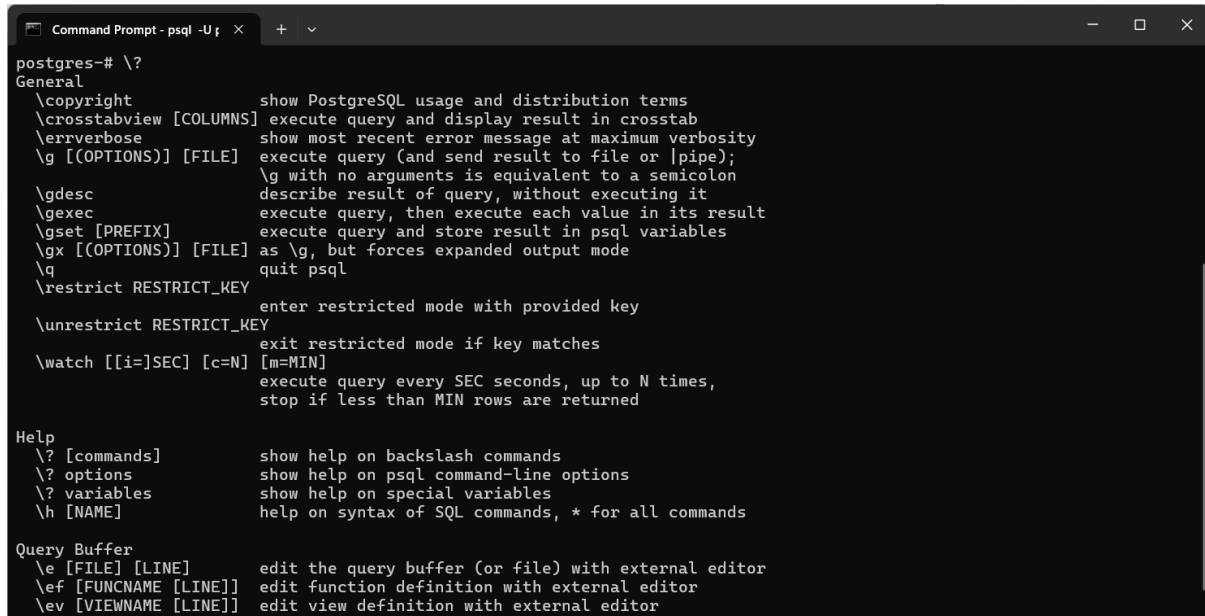
(3) begin a transaction using `conn.execute("BEGIN")` or commit a transaction using `conn.commit()` or rollback using `conn.rollback()` if it fails (Atomicity principle)

...

## PostgreSQL

Follow the instructions on <https://www.enterprisedb.com/docs/supported-open-source/postgresql/installing/windows/> (or Linux depending on your machine).

For help from the CLI, type \?



```
Command Prompt - psql -U f x + v
postgres=# \?
General
\copyright          show PostgreSQL usage and distribution terms
\crosstabview [COLUMNS] execute query and display result in crosstab
\errverbose         show most recent error message at maximum verbosity
\g [(OPTIONS)] [FILE] execute query (and send result to file or |pipe);
                    \g with no arguments is equivalent to a semicolon
\gdesc             describe result of query, without executing it
\gexec             execute query, then execute each value in its result
\gset [PREFIX]     execute query and store result in psql variables
\gx [(OPTIONS)] [FILE] as \g, but forces expanded output mode
\q                quit psql
\restrict RESTRICT_KEY enter restricted mode with provided key
\unrestrict RESTRICT_KEY exit restricted mode if key matches
\watch [(i=)SEC] [c=N] [m=MIN] execute query every SEC seconds, up to N times,
                    stop if less than MIN rows are returned

Help
\? [commands]      show help on backslash commands
\? options         show help on psql command-line options
\? variables       show help on special variables
\h [NAME]          help on syntax of SQL commands, * for all commands

Query Buffer
\e [FILE] [LINE]   edit the query buffer (or file) with external editor
\ef [FUNCNAME] [LINE] edit function definition with external editor
\ev [VIEWNAME] [LINE] edit view definition with external editor
```

```
C:\users\sarah>psql -U postgres
```

Password for user postgres:

```
postgres=# CREATE DATABASE firstPostgresDB ;
```

```
postgres=# \l
```

```
postgres=# \c firstpostgresdb
```

You are now connected to database "firstpostgresdb" as user "postgres".

```
firstpostgresdb=# \i 'C:/path/to/your/SQLfile/eCommercePostgreSQL.sql'
```

```
CREATE TABLE
```

```
CREATE TABLE
```

```
CREATE TABLE
```

```
CREATE TABLE
```

```
CREATE TABLE
```

```
CREATE INDEX
```

```
CREATE INDEX
```

CREATE INDEX

CREATE PROCEDURE

CREATE PROCEDURE

INSERT 0 6

INSERT 0 2

INSERT 0 2

firstpostgresdb=# SELECT \* FROM products;

| product_id | name                | description              | price  | stock |
|------------|---------------------|--------------------------|--------|-------|
| 1          | Laptop Pro 15       | High-performance laptop  | 999.99 | 10    |
| 2          | Wireless Mouse      | Ergonomic wireless mouse | 29.99  | 50    |
| 3          | USB-C Hub           | 7-in-1 USB-C hub         | 49.99  | 30    |
| 4          | Mechanical Keyboard | RGB mechanical keyboard  | 89.99  | 20    |
| 5          | Monitor 27"         | 4K IPS display           | 399.99 | 8     |
| 6          | Webcam HD           | 1080p webcam with mic    | 59.99  | 25    |

(6 rows)

postgres=# \conninfo

| Connection Information |           |
|------------------------|-----------|
| Parameter              | Value     |
| Database               | postgres  |
| Client User            | postgres  |
| Host                   | localhost |
| Host Address           | ::1       |
| Server Port            | 5432      |
| Options                |           |
| Protocol Version       | 3.0       |
| Password Used          | true      |
| GSSAPI Authenticated   | false     |
| Backend PID            | 129444    |
| SSL Connection         | false     |
| Superuser              | on        |
| Hot Standby            | off       |

(13 rows)

Note: **Ensure** that you use the correct database name. The one created on PostgreSQL is the name passed to python.



**Note:**

- There are differences in the syntax between the different SQL languages. For example:  
MySQL uses `AUTO_INCREMENT`, while SQLite uses `AUTOINCREMENT`. You will get an error message if you do not use the write syntax.
- Also if from the CLI of PostgreSQL, when importing for Windows users: Use **forward slashes /** for the file path, even on Windows, or the command might fail. Wrap the path in single quotes.
- It is assumed the server is run on the same device (localhost with IP address 127.0.0.1), but if it is running on a separate device, use the appropriate connection information.

## **SQL practice**

Given the database, how to express the following queries in SQL.

Given the database what is the resulting output of the following queries.